Completed Gaussian quadratic products: sharp fourth moments and contraction tests

Project Stardust
A companion to a frozen verified library

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Abstract

Centered Gaussian quadratic forms extend from finite matrices to the complete second Wiener chaos. We give complete proofs of the sharp dimension-independent fourth-moment bounds, bounded multiplication into Gaussian L^2, its exact norm, and the degree-zero/two/four product decomposition. A nonzero completed quadratic has strictly larger fourth moment than a Gaussian of the same variance, although the lower universal coefficient is not attained. Polarization gives nonnegative covariance of quadratic squares with exact contraction conditions for equality. For finite symmetric matrices these conditions reduce to matrix annihilation, yielding the classical independence test. The completion and its source/operator normalization are made explicit. A centered first/second-chaos mixture has negative square covariance, delimiting the homogeneous-chaos hypothesis. Combined tensor symmetrization retains a double-skew determinant whose actual transported law is explicitly Laplace. A two-source product has a sharper lower fourth-moment coefficient of nine, and an explicit anticommuting pair delimits incomplete independence tests. Full second-chaos prediction compresses both completed matrix coordinates; its exact error retains the mixed observed/unobserved block. Classical quadratic-form theory is credited throughout; no new independence criterion or fourth-moment phenomenon is claimed.

1 The actual Gaussian space and the main statements

Let $(G_i)_{i\geq 0}$ be independent standard real Gaussian variables on their product probability space. Every L^p space below uses this positive probability law. Let $\mathcal{H}_2$ be the closed subspace of L^2 generated by the centered quadratic monomials

$$G_i^2 - 1, \qquad G_iG_j \quad (i < j).$$

This is the second Wiener chaos. Orthogonality of the tensor Hermite polynomials also defines the closed degree-k spaces $\mathcal{H}_k$ and their orthogonal projections P_k. We use the probabilists' normalization $H_2(x) = x^2 - 1$, $H_4(x) = x^4 - 6x^2 + 3$. Here $\mathcal{H}_0$ is the constant space and $P_0F = (\mathbb{E}F)1$.

Theorem 1.1. *Every $F \in \mathcal{H}_2$ belongs to L^4. If $F \neq 0$, then*

$$3(\mathbb{E}F^2)^2 < \mathbb{E}F^4 \leq 15(\mathbb{E}F^2)^2. \tag{1}$$

The upper coefficient is attained. The lower coefficient 3 is optimal over the entire space, but equality there forces $F = 0$. Multiplication is a bounded bilinear map

$$(F, G) \longmapsto FG : \mathcal{H}_2 \times \mathcal{H}_2 \to L^2, \qquad \|FG\|_2 \leq \sqrt{15}\|F\|_2\|G\|_2,$$

whose norm is exactly $\sqrt{15}$*. Its range lies in* $\mathcal{H}_0 \oplus \mathcal{H}_2 \oplus \mathcal{H}_4$*, rather than in* $\mathcal{H}_2$ *alone. The exact fourth-cumulant identity is*

$$\mathbb{E}F^4 - 3(\mathbb{E}F^2)^2 = \frac{3}{2}\|P_2(F^2)\|_2^2. \tag{2}$$

For all $F, G \in \mathcal{H}_2$*,*

$$\begin{aligned}\operatorname{Cov}(F^2, G^2) &= 2\langle F, G\rangle^2 + \frac{1}{2}\langle P_2(F^2), P_2(G^2)\rangle + \|P_2(FG)\|_2^2 \\ &\geq 2\langle F, G\rangle^2 \geq 0.\end{aligned} \tag{3}$$

Every term on the first line's right side is nonnegative. In particular equality $\operatorname{Cov}(F^2, G^2) = 0$ *holds precisely when all three displayed terms vanish.*

The assertion about the middle contraction pairing is special to quadratic chaos. It is not a generic assertion that inner products of arbitrary orthogonal projections are nonnegative. Likewise, uncorrelated quadratic observables need not be independent.

Lancaster's primary quadratic-form work [1] identifies cumulants with matrix traces. Ogawa's primary account [2] states the classical centered Gaussian matrix-annihilation independence criterion. The present paper reconstructs the exact completed-space mechanisms in a verified library; it does not claim priority for those classical results. Primary full-text access for Lancaster's detailed hypotheses remains incomplete.

2 Finite matrices, observable normalization and moments

For a real symmetric $n \times n$ matrix A define

$$Q_A = \sum_{i,j<n} A_{ij}(G_iG_j - \delta_{ij}) = G^T AG - \operatorname{tr} A. \tag{4}$$

If a matrix is not symmetric, its observable equals that of $S(A) = (A + A^T)/2$: transpose the double sum to see that the antisymmetric part vanishes. Thus the source observable identifies matrices differing by a skew-symmetric matrix. We first work on the symmetric representative to keep its norm nondegenerate.

The generating function of independent standard Gaussian coordinates is $\mathbb{E}e^{\langle u,G\rangle} = e^{\|u\|^2/2}$. Expanding it to degree four gives

$$\mathbb{E}G_iG_jG_kG_l = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}.$$

Consequently,

$$\mathbb{E}Q_A = 0, \qquad \langle Q_A, Q_B\rangle = 2\sum_{i,j} A_{ij}B_{ij}, \qquad \mathbb{E}Q_A^2 = 2\|A\|_{\mathrm{HS}}^2. \tag{5}$$

The Hilbert–Schmidt norm here is the Frobenius norm on finite matrices, $\|A\|_{\mathrm{HS}}^2 = \sum_{i,j} A_{ij}^2$.

An orthogonal change of coordinates preserves the standard Gaussian law, since it preserves its characteristic function $e^{-\|u\|^2/2}$. Diagonalize A by a real orthogonal matrix. If its eigenvalues are a_i, then Q_A has the law of $\sum_i a_i(G_i^2 - 1)$. Set $U_i = G_i^2 - 1$. The elementary normal moments $\mathbb{E}G^{2k} = (2k-1)!!$ follow by Gaussian integration by parts, and give

$$\mathbb{E}U_i = 0, \qquad \mathbb{E}U_i^2 = 2, \qquad \mathbb{E}U_i^4 = 60.$$

In the fourth power of the independent sum, only the four-equal and two-pair index patterns survive. Thus

$$\begin{aligned}\mathbb{E}Q_A^4 &= 60\sum_i a_i^4 + 24\sum_{i<j} a_i^2 a_j^2 \\ &= 12\Big(\sum_i a_i^2\Big)^2 + 48\sum_i a_i^4.\end{aligned} \tag{6}$$

Since $\sum_i a_i^4 \le (\sum_i a_i^2)^2$, this proves the finite upper bound $\mathbb{E}Q_A^4 \le 15(\mathbb{E}Q_A^2)^2$. Also

$$\mathbb{E}Q_A^4 - 3(\mathbb{E}Q_A^2)^2 = 48\|A^2\|_{\mathrm{HS}}^2, \tag{7}$$

because the eigenvalues of A^2 are a_i^2. No simultaneous diagonalization of two unrelated symmetric matrices is used.

3 The full second chaos and its original source field

Let $\mathcal{S}_2$ be the real Hilbert space of symmetric square-summable matrices $A = (A_{ij})_{i,j\ge 0}$, with its Hilbert–Schmidt norm. Let $A^{(n)}$ retain its first n rows and columns and set all other entries to zero. Then $A^{(n)} \to A$ in that norm. By (5), the actual random variables $Q_{A^{(n)}}$ are Cauchy in L^2. Define

$$Q_A = \lim_{n\to\infty} Q_{A^{(n)}} \quad \text{in } L^2.$$

The resulting map satisfies

$$\|Q_A\|_2^2 = 2\|A\|_{\mathrm{HS}}^2. \tag{8}$$

It is onto $\mathcal{H}_2$: its range is closed by this isometry and contains every centered quadratic monomial. Conversely, every finite-box observable lies in the closed span defining $\mathcal{H}_2$. This proves both inclusions and uniqueness of the symmetric matrix. The definition does not require $\sum_i A_{ii}$ to converge. The formal expression $G^T AG - \operatorname{tr} A$ is only the finite formula; the completed centered quadratic is an L^2 limit, not an unjustified subtraction of two infinite quantities.

The same matrix has an actual Gaussian linear source field. Let $H = \ell^2(\mathbb{N})$ with orthonormal basis (e_i) and define

$$V_A = \lim_{n\to\infty} \sum_{i,j<n} A_{ij} e_i G_j \quad \text{in } L^2(\Omega; H).$$

Orthonormality gives $\|V_A\|^2 = \|A\|_{\mathrm{HS}}^2$. On a finite linear field $V = (V_i)$, the Gaussian divergence is

$$\delta V = \sum_i (G_i V_i - \partial_i V_i).$$

Hence $\delta V_A = Q_{S(A)}$ for finite matrices, and $\|\delta V_A\|_2^2 = 2\|S(A)\|_{\mathrm{HS}}^2$. It extends boundedly to the completion of linear fields, with norm $\sqrt{2}$, and agrees with the original closed divergence there. To check the last assertion, for each smooth cylindrical test f, finite Gaussian integration by parts gives

$$\mathbb{E}(f\,\delta V_{A^{(n)}}) = \mathbb{E}\langle Df, V_{A^{(n)}}\rangle_H.$$

Pass to the two L^2 limits. This is precisely the adjoint-domain identity defining the original divergence, with a unique output. Thus bounded completion is compatible with the original operator, rather than defining a second unrelated divergence.

For symmetric A, the recovery field of Q_A is V_A and has energy $\|V_A\|^2 = \frac{1}{2}\|Q_A\|_2^2$. On finite boxes this is also $\frac{1}{2}DQ_A$, since $DQ_A = 2V_A$. The source's recovery and symmetric-field unitary extend this identity to the full second chaos. In this paper it suffices to retain the explicit coefficient isometry and finite-box approximations. The matrix notation is a human coordinate realization of those source constructions.

4 Full quadratic prediction is a two-coordinate compression

Let $M \subseteq H$ be any closed subspace, P its orthogonal projection and $R = I - P$. Put $W(h) = \sum_i h_i G_i$, with convergence in L^2, and let $\mathcal{F}_M = \sigma(W(h) : h \in M)$, completed under the original probability law. No finite-rank hypothesis on P is imposed. A symmetric square-summable matrix defines a self-adjoint Hilbert–Schmidt operator A on H; its centered observable is the actual Q_A already constructed above.

Theorem 4.1. *For every such A,*

$$\mathbb{E}[Q_A \mid \mathcal{F}_M] = Q_{PAP}, \qquad \|\mathbb{E}[Q_A \mid \mathcal{F}_M]\|_2^2 = 2\|PAP\|_{\mathrm{HS}}^2, \tag{9}$$
$$\|Q_A - \mathbb{E}[Q_A \mid \mathcal{F}_M]\|_2^2 = 4\|RAP\|_{\mathrm{HS}}^2 + 2\|RAR\|_{\mathrm{HS}}^2.$$

For every $Y \in L^2(\mathcal{F}_M)$, the exact error decomposition is

$$\|Q_A - Y\|_2^2 = \|Q_A - Q_{PAP}\|_2^2 + \|Q_{PAP} - Y\|_2^2.$$

Thus Q_{PAP} is the unique best predictor even among nonlinear square-integrable functions of all observations. Zero prediction error holds exactly when $A = PAP$.

Proof. For finite lists $h_1, \dots, h_k$, the joint Gaussian characteristic function is $\exp(-\| \sum_j t_j h_j\|^2/2)$. Orthogonality of M and $M^\perp$ therefore factors every mixed characteristic function. The two generated sigma algebras are independent: finite cylinder events are independent, and two applications of the monotone-class theorem extend this to the generated sigma algebras. Null completions preserve independence.

Split $h = Ph + Rh$. Independence, zero means and covariance isometry give

$$\mathbb{E}[W(a)W(b) - \langle a, b\rangle \mid \mathcal{F}_M] = W(Pa)W(Pb) - \langle Pa, Pb\rangle.$$

For example the unobserved product contributes the deterministic $\langle Ra, Rb\rangle$, which cancels that part of the centering; each mixed product has conditional mean zero. Symmetric finite-rank tensors span a dense subspace of the symmetric Hilbert–Schmidt operators. Summing this identity proves the prediction formula there. For an arbitrary A, choose symmetric finite-rank $A_n \to A$ in Hilbert–Schmidt norm. Conditional expectation is an L^2 contraction, $\|P(A_n - A)P\|_{\mathrm{HS}} \le \|A_n - A\|_{\mathrm{HS}}$, and (8) passes the identity to the actual limits.

The four blocks of an operator relative to $H = M \oplus M^\perp$ are orthogonal in Hilbert–Schmidt inner product. Symmetry gives $\|PAR\|_{\mathrm{HS}} = \|RAP\|_{\mathrm{HS}}$, so

$$\|A\|_{\mathrm{HS}}^2 = \|PAP\|_{\mathrm{HS}}^2 + 2\|RAP\|_{\mathrm{HS}}^2 + \|RAR\|_{\mathrm{HS}}^2.$$

This follows directly by summing squared matrix entries in orthonormal bases of the two closed subspaces, including empty or finite bases. The isometry gives both norm formulas. Conditional-expectation duality makes $Q_A - Q_{PAP}$ orthogonal to $L^2(\mathcal{F}_M)$, proving the error decomposition and uniqueness. Finally its error vanishes exactly when the two remaining blocks vanish, equivalently $A = PAP$. □

The source represents the recovery field of Q_A by its square-summable matrix coordinates. Applying the source projection in the direction coordinate and conditional expectation in the random coordinate sends V_A to V_{PAP}. This can be verified on $aW(b)$, where it gives $(Pa)W(Pb)$,

and then by bounded completion. Compression commutes with transposition because $P = P^*$. It consequently preserves symmetric fields. The completed divergence identity

$$\delta V_{PAP} = Q_{PAP} = \mathbb{E}[\delta V_A \mid \mathcal{F}_M]$$

retains the original closed divergence, not a new formal matrix operator. In coordinates its exact prediction energy is

$$2 \sum_{i,j \geq 0} \langle (Pe_i) W (Pe_j), V_A \rangle^2_{L^2(\Omega;H)} = \|\mathbb{E}[Q_A \mid \mathcal{F}_M]\|_2^2.$$

Indeed the inner product is $\langle Pe_i, APe_j \rangle$, the (i,j) coefficient of PAP; Parseval gives the sum. The source proves this for every completed second-chaos target, not only finite quadratic polynomials. The least-error and block-error formulations above are explicit human Hilbert-space consequences.

Conditional mean zero does not make the residual independent

One can say more about the error without discarding its dependence. Let $C = RAP$ and $D = RAR$. The observed H-valued Gaussian field

$$U_C = \sum_j C m_j W(m_j) \quad \text{in } L^2(\Omega; H)$$

uses an orthonormal basis (m_j) of M. It is well defined since $\sum_j \|Cm_j\|^2 = \|C\|^2_{\mathrm{HS}}$ and is $\mathcal{F}_M$-measurable. The following additional identity is a human conditional-moment deduction, not a new mapped formal conditional-variance declaration:

$$\mathbb{E}[(Q_A - Q_{PAP})^2 \mid \mathcal{F}_M] = 4\|U_C\|_H^2 + 2\|D\|^2_{\mathrm{HS}} \quad \text{in } L^1. \tag{10}$$

To prove it, first use finite adapted bases of M and $M^\perp$. Writing their independent standard coordinates as Y and Z, the residual is $2Z^T CY + Q_D(Z)$. Conditional on Y, its two terms are uncorrelated: the mixed third centered Gaussian moments vanish. Their variances are $4\|CY\|^2$ and $2\|D\|^2_{\mathrm{HS}}$. This proves the finite formula. Truncate all three operator blocks in these adapted bases. Hilbert–Schmidt convergence gives residual convergence in L^2, hence squared-residual convergence in L^1 by Cauchy–Schwarz. Also $U_{C_n} \to U_C$ in $L^2(\Omega; H)$, so its squared norm converges in L^1 by the same inequality. Conditional expectation is an L^1 contraction. These facts pass the finite identity to (10), without a pointwise infinite quadratic expression or an H-valued standard Gaussian.

For a concrete warning take $M = \mathrm{span}\{e_0\}$ and $A_{01} = A_{10} = 1$, all other entries zero. Then $Q_A = 2G_0G_1$, $PAP = RAR = 0$ and $\mathbb{E}[Q_A \mid G_0] = 0$, but

$$\mathbb{E}[Q_A^2 \mid G_0] = 4G_0^2$$

is nonconstant. The residual is therefore not independent of the observations. This also shows why the mixed block contributes twice in the Hilbert–Schmidt error and cannot be omitted.

Gaussian conditioning as second quantization is classical. Alpay–Kipnis [6], Section 2, printed page 5, identifies conditional expectation with the projected chaos operator before equation (2.8). Our proof uses this mechanism directly on quadratic tensors and proves its completion. Only the conditioning identity is used from that comparison; no implication from conditional mean zero to independence of a mixed chaos is assumed.

5 Strong fourth-moment completion and bounded multiplication

The finite upper bound applies to the symmetric difference of two box matrices. Therefore

$$\|Q_{A^{(n)}} - Q_{A^{(m)}}\|_4 \le 15^{1/4}\|Q_{A^{(n)}} - Q_{A^{(m)}}\|_2.$$

The sequence is Cauchy in L^4. Its L^4 limit has the same L^2 class as Q_A, because the probability measure gives $\|X\|_2 \le \|X\|_4$. Completeness and uniqueness identify the limits. Hence every $F \in \mathcal{H}_2$ belongs to L^4 and

$$\|F\|_4 \le 15^{1/4}\|F\|_2. \tag{11}$$

This also proves the upper bound in (1) on the full space. No continuity of squaring on unrestricted L^2 is used.

By Hölder and (11),

$$\|FG\|_2 \le \|F\|_4\|G\|_4 \le \sqrt{15}\|F\|_2\|G\|_2.$$

The product is the actual almost-everywhere pointwise product, since both factors have L^4 representatives. For box approximations,

$$\|F_nG_n - FG\|_2 \le \|F_n - F\|_4\|G_n\|_4 + \|F\|_4\|G_n - G\|_4 \longrightarrow 0.$$

In particular $F_n^2 \to F^2$ strongly in L^2. Bilinearity on equivalence classes is inherited from pointwise multiplication.

Take $F = G = G_0^2 - 1$. Its variance is 2 and fourth moment is 60. The quotient $\|F^2\|_2/(\|F\|_2\|F\|_2)$ is $\sqrt{60}/2 = \sqrt{15}$. Thus the bilinear norm bound is exact, and the upper fourth-moment coefficient is attained. The image need not be quadratic: $(G_0^2 - 1)^2$ has a nonzero fourth-chaos component, identified below.

6 Degree projections and the completed cumulant

For finite symmetric matrices the degree-two product component is

$$P_2(Q_AQ_B) = 2Q_{AB+BA}, \qquad P_2(Q_A^2) = 4Q_{A^2}. \tag{12}$$

One direct verification uses the degree-two product rule

$$\begin{aligned}(G_iG_j - \delta_{ij})(G_kG_l - \delta_{kl}) &\\ = \text{degree-four Wick term} + \delta_{ik}(G_jG_l - \delta_{jl}) + \delta_{il}(G_jG_k - \delta_{jk}) &\\ + \delta_{jk}(G_iG_l - \delta_{il}) + \delta_{jl}(G_iG_k - \delta_{ik}) + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}.&\end{aligned}$$

This identity follows by expanding the multivariate Hermite generating function $\exp(\langle t, G\rangle - \|t\|^2/2)$ in its two pairs of variables. Equivalently, subtract the listed lower degree terms from the product and check its orthogonality to degrees zero through three by Gaussian moments. Summing with coefficients $A_{ij}B_{kl}$ gives (12). It gives $P_0(FG) = \langle F, G\rangle 1$ as well.

For finite observables the product has only degrees $0, 2, 4$. The sum of those three mutually orthogonal closed subspaces is closed: if a sequence in their sum is Cauchy, each projected component is Cauchy and converges in its closed subspace. Strong product convergence therefore gives, for the full second chaos,

$$FG = P_0(FG) + P_2(FG) + P_4(FG). \tag{13}$$

All other degree projections vanish. This is a decomposition in the original Gaussian L^2, not a new multiplication internal to $\mathcal{H}_2$.

For a finite symmetric A, (12) and (5) give

$$\|P_2(Q_A^2)\|_2^2 = 32\|A^2\|_{\mathrm{HS}}^2.$$

Combining this with (7) proves (2) on boxes. Strong L^4 approximation of F and strong L^2 approximation of its square, followed by the bounded orthogonal projection P_2, pass every term to the limit. This proves (2) on the entire second chaos. The argument uses finite-dimensional spectral theory only for the finite fourth-moment calculation; no infinite spectral expansion or trace-class hypothesis on the original quadratic matrix is needed.

Orthogonality in (13) also yields

$$\|P_4(F^2)\|_2^2 = 2\|F\|_2^4 + \frac{1}{2}\|P_2(F^2)\|_2^2. \tag{14}$$

Indeed, $\|F^2\|_2^2 = \|F\|_2^4 + \|P_2(F^2)\|_2^2 + \|P_4(F^2)\|_2^2$, and (2) identifies the left side. In particular,

$$\|P_2(F^2)\|_2^2 \le 8\|F\|_2^4, \qquad 2\|F\|_2^4 \le \|P_4(F^2)\|_2^2 \le 6\|F\|_2^4.$$

These are exact degree-energy consequences, not pointwise positivity statements about the projected functions.

7 Strictness without a uniform gap above three

We prove that the second-degree square contraction of a nonzero completed quadratic cannot vanish. This strictness does not follow merely by taking limits of positive finite cumulants, since positive numbers can tend to zero.

Represent $F = Q_A$ by its unique symmetric square-summable matrix. For fixed i, j, its finite-box contraction satisfies, for $n > i, j$,

$$\frac{1}{4}\big[\text{recovery of } P_2(Q_{A^{(n)}}^2)\big]_{ii} = \sum_{k<n} A_{ik}^2 \ge A_{ij}^2.$$

Here recovery sends Q_C to V_C; an individual source coefficient is a bounded linear functional on the completed field. Recovery itself is bounded on the second chaos by the energy identity in Section 3. Since $Q_{A^{(n)}}^2 \to F^2$ in L^2, the coefficient inequality passes to the limit. If $P_2(F^2) = 0$, its recovered diagonal coefficient is zero. Thus $A_{ij}^2 = 0$ for every fixed pair i, j, so $A = 0$ and $F = 0$. This proves

$$F \neq 0 \implies P_2(F^2) \neq 0.$$

Equation (2) now gives the strict lower inequality in Theorem 1.1. Since every $F \in \mathcal{H}_2$ is centered, no nonzero such variable has a Gaussian law: a centered Gaussian of variance $\mathbb{E}F^2$ has fourth moment exactly $3(\mathbb{E}F^2)^2$.

Nevertheless there is no larger universal lower coefficient. For $n \ge 1$, set $T_n = \sum_{i=0}^{n-1}(G_i^2 - 1)$. The finite moment calculation gives

$$\mathbb{E}T_n^2 = 2n, \qquad \mathbb{E}T_n^4 = 12n^2 + 48n, \qquad \frac{\mathbb{E}T_n^4}{(\mathbb{E}T_n^2)^2} = 3 + \frac{12}{n} \longrightarrow 3.$$

If a constant c were a lower fourth-moment coefficient for every quadratic in the completed space, these explicit tests would give $c \le 3 + 12/n$ for every n, hence $c \le 3$. Thus strictness for each nonzero target and optimality of the nonattained lower coefficient coexist. No distributional limit theorem for T_n is needed or asserted in this proof.

8 Mixed square covariance by polarization

We next prove the exact mixed identity (3). Set $U = P_2(F^2)$, $V = P_2(G^2)$ and $W = P_2(FG)$. Apply (2) to $F + G$ and $F - G$. Subtract twice its versions for F and G from the sum of those two identities. The fourth-moment contribution is $12\mathbb{E}(F^2G^2)$. Writing $a = \|F\|_2^2$, $b = \|G\|_2^2$, $c = \langle F, G\rangle$, the Gaussian subtraction contributes $-12ab - 24c^2$. On the contraction side,

$$P_2((F \pm G)^2) = U + V \pm 2W.$$

The parallelogram identity then gives the remaining contraction contribution $6\langle U, V\rangle + 12\|W\|_2^2$. Dividing by 12 proves (3). All products and fourth moments exist by Section 4, so polarization is valid on the completed space.

It remains to show $\langle U, V\rangle \geq 0$. For finite symmetric A, B, (12) gives

$$\langle P_2(Q_A^2), P_2(Q_B^2)\rangle = 32\,\mathrm{tr}(A^2B^2) = 32\|AB\|_{\mathrm{HS}}^2 \geq 0. \tag{15}$$

The matrix identity follows from $\|AB\|_{\mathrm{HS}}^2 = \mathrm{tr}((AB)^TAB) = \mathrm{tr}(BA^2B) = \mathrm{tr}(A^2B^2)$, using symmetry and cyclicity of finite traces. Approximate F, G by their finite boxes. Strong square convergence and boundedness of P_2 make the two projected squares converge strongly, so their inner products converge. Nonnegativity therefore passes to arbitrary $F, G \in \mathcal{H}_2$. This proves the lower bounds and the exact equality conditions in Theorem 1.1.

9 Finite-matrix annihilation and the independence boundary

For finite symmetric A, B, the preceding identities specialize to

$$\mathrm{Cov}(Q_A^2, Q_B^2) = 8(\mathrm{tr}\,AB)^2 + 16\|AB\|_{\mathrm{HS}}^2 + 8\|AB + BA\|_{\mathrm{HS}}^2. \tag{16}$$

Here $\langle Q_A, Q_B\rangle = 2\,\mathrm{tr}\,AB$ and $P_2(Q_AQ_B) = 2Q_{AB+BA}$, while (15) gives the middle term. Every term on the right is nonnegative. It follows that

$$\mathrm{Cov}(Q_A^2, Q_B^2) = 0 \quad \Longleftrightarrow \quad AB = 0. \tag{17}$$

The reverse implication also uses $BA = (AB)^T = 0$. If Q_A, Q_B are independent, their squares are independent and their finite fourth moments imply factorization of the product of squares. Equation (17) then forces $AB = 0$. This is the source-mapped necessary independence test.

For completeness, the classical converse is an elementary finite-dimensional corollary, rather than an additional formal declaration asserted in the source map. If $AB = 0$, symmetry gives $BA = 0$ and $\mathrm{ran}\,A \perp \mathrm{ran}\,B$: $\langle Ax, By\rangle = \langle x, ABy\rangle = 0$. Choose an orthonormal basis adapted to these two orthogonal ranges and their joint orthogonal complement. A symmetric matrix annihilates the orthogonal complement of its range; hence Q_A depends only on the Gaussian coordinates in the first range, and Q_B only on those in the second. Those coordinate vectors are independent, so the two observables are independent. Thus (17) is also equivalent to independence in this finite centered Gaussian setting. This is classical quadratic-form theory, not a newly claimed criterion.

Orthogonality of the targets is weaker. The explicit source example in Section 11 also has vanishing mixed degree-two contraction, while its square-contraction pairing remains positive.

10 Sharp two-source product bounds

The general lower coefficient 3 is not the sharp coefficient on the smaller class of centered products of two linear Gaussian sources. Let W be the actual isonormal embedding of the real source Hilbert space and set

$$X = W(u), \quad Y = W(v), \quad a = \|u\|^2, \quad b = \|v\|^2, \quad c = \langle u, v\rangle, \quad A = ab, \quad Z = XY - c.$$

These definitions allow arbitrary completed source vectors, not only finitely supported vectors. Their two-dimensional joint Gaussian law has moment generating function

$$\mathbb{E}e^{sX+tY} = \exp\left(\frac{as^2 + 2cst + bt^2}{2}\right).$$

Indeed every linear combination is the embedded source $W(su + tv)$. This identity also covers singular covariance matrices. All polynomial moments are finite: Cauchy–Schwarz bounds each mixed absolute moment by the corresponding one-variable Gaussian moments.

Proposition 10.1. *The actual centered product satisfies*

$$\begin{aligned} \mathbb{E}Z^2 &= A + c^2, \\ \mathbb{E}Z^4 &= 9A^2 + 42Ac^2 + 9c^4. \end{aligned} \tag{18}$$

Consequently

$$9(\mathbb{E}Z^2)^2 \le \mathbb{E}Z^4 \le 15(\mathbb{E}Z^2)^2, \qquad 15(\mathbb{E}Z^2)^2 - \mathbb{E}Z^4 = 6(A - c^2)^2. \tag{19}$$

If $Z \neq 0$ in L^2, lower equality holds exactly when $c = 0$; upper equality holds exactly when $A = c^2$. Thus both constants are attained by nonzero targets. The latter equality is precisely equality in the source Cauchy–Schwarz inequality.

Proof. Expanding the displayed exponential, the coefficient of $s^k t^k$ is the sum over nonnegative integers i, j, l with $2i + j = k$ and $j + 2l = k$ of

$$\frac{(a/2)^i c^j (b/2)^l}{i!j!l!}.$$

Multiply by $(k!)^2$. For $k = 1, 2, 3, 4$ this gives respectively

$$\mathbb{E}XY = c, \quad \mathbb{E}X^2Y^2 = A + 2c^2, \quad \mathbb{E}X^3Y^3 = 9Ac + 6c^3,$$

$$\mathbb{E}X^4Y^4 = 9A^2 + 72Ac^2 + 24c^4.$$

Expanding $(XY - c)^2$ and $(XY - c)^4$ proves (18). Subtracting nine times the squared variance gives $24Ac^2 \ge 0$; subtracting the fourth moment from fifteen times the squared variance gives the stated squared Gram determinant. For a nonzero target, $A > 0$: if $a = 0$ or $b = 0$, the corresponding centered Gaussian is zero almost surely and so is Z. Thus lower equality is equivalent to $c = 0$, while upper equality is equivalent to $A = c^2$. Orthogonal nonzero vectors give the first case, and $u = v \neq 0$ gives the second. □

For $A > 0$ put $r = c^2/A \in [0,1]$. A further elementary human reorganization of the moment identity is

$$\frac{\mathbb{E}Z^4}{(\mathbb{E}Z^2)^2} = 9 + \frac{24r}{(1+r)^2}.$$

Its derivative with respect to r is $24(1-r)/(1+r)^3 \geq 0$. This describes the exact interpolation between an orthogonal product and a rank-one square. It is not a lower bound of nine for the whole completed second chaos; the many-coordinate examples in the earlier section still approach three. The formal module `GaussianMixedQuadraticFourthMoment` supplies `mixed_fourth_moment`, `mixed_fourth_upper_deficit`, `mixed_fourth_upper_equality_iff` and `mixed_fourth_lower_bound` retain the actual source targets. The lower equality characterization and scalar monotonicity just proved are explicit human consequences. These Gaussian product moments are classical: Gaunt [5], Section 2.7, gives the same central moments and kurtosis after normalizing the variances and setting the number of product summands to one.

11 A vanishing mixed contraction does not give independence

The three conditions in the square-covariance identity must be kept together. Let X, Y be two actual independent standard source coordinates and take

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad F = X^2 - Y^2, \quad G = 2XY.$$

Both are centered members of $\mathcal{H}_2$. Matrix multiplication gives $AB = -BA = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, so their anticommutator vanishes but their product does not. In particular

$$\langle F, G \rangle = 0, \qquad \mathbb{E}F^2 = \mathbb{E}G^2 = 4.$$

The first equality follows from odd one-coordinate Gaussian moments; the variances follow from $\mathbb{E}X^4 = \mathbb{E}Y^4 = 3$.

Using $H_3(x) = x^3 - 3x$, cancellation of the two degree-two terms gives

$$FG = 2H_3(X)Y - 2XH_3(Y) \in \mathcal{H}_4, \qquad P_2(FG) = 0.$$

Yet the square contractions are

$$P_2(F^2) = P_2(G^2) = 4\big(H_2(X) + H_2(Y)\big).$$

For the first identity expand $F^2 = X^4 + Y^4 - 2X^2Y^2$ with $x^4 = H_4(x) + 6H_2(x) + 3$ and $x^2 = H_2(x) + 1$. For the second use $G^2 = 4X^2Y^2$. Since $H_2(X)$ and $H_2(Y)$ are orthogonal and each has squared norm two, their contraction pairing is 64. Therefore the earlier polarization identity yields

$$\operatorname{Cov}(F^2, G^2) = \tfrac{1}{2}\, 64 = 32 > 0. \tag{20}$$

One can also check this without any chaos projection:

$$\mathbb{E}F^2G^2 = 4\mathbb{E}(X^6Y^2 - 2X^4Y^4 + X^2Y^6) = 4(15 - 18 + 15) = 48,$$

and subtracting $4 \cdot 4$ gives 32. Independence would factor these square moments, a contradiction. This is a classical explicit calibration of the matrix-annihilation criterion, not a replacement criterion.

Even orthogonality together with $P_2(FG) = 0$ omits the nonnegative square-contraction pairing in (3). The pointwise algebraic identity

$$F^2 + G^2 = (X^2 + Y^2)^2$$

is consistent with this dependence, but the positive covariance above is the actual proof. All these source identifications and constants are recorded in `GaussianQuadraticAnticommutingExample`; no assumption that nonlinear Gaussian targets are jointly Gaussian has been made.

12 Centered mixed chaoses need not have associated squares

The restriction to homogeneous second chaos in (3) cannot be replaced by centering alone, even for degree-two polynomials of one Gaussian coordinate. Let $Z = G_0$ and, for $a \in \mathbb{R}$, define the actual random variables

$$F_a = Z + a(Z^2 - 1), \qquad G_a = Z - a(Z^2 - 1).$$

They lie in $\mathcal{H}_1 \oplus \mathcal{H}_2$, are centered, and have moments of every order. Orthogonality of H_1 and H_2 gives

$$\|F_a\|_2^2 = \|G_a\|_2^2 = 1 + 2a^2, \qquad \langle F_a, G_a \rangle = 1 - 2a^2. \tag{21}$$

There is an exact product identity

$$F_a G_a = (1 - 2a^2)H_0(Z) + (1 - 4a^2)H_2(Z) - a^2 H_4(Z). \tag{22}$$

Indeed the left side is $Z^2 - a^2(Z^2 - 1)^2$, while $Z^2 = H_2 + 1$ and $(Z^2 - 1)^2 = H_4 + 4H_2 + 2$. No completion or formal product is involved: these are polynomial equalities pointwise for every real Z. Gaussian Hermite orthogonality, with squared norms $1, 2, 24$ in degrees $0, 2, 4$, yields

$$\begin{aligned} \mathbb{E}(F_a^2 G_a^2) &= (1 - 2a^2)^2 + 2(1 - 4a^2)^2 + 24a^4 = 3 - 20a^2 + 60a^4, \\ \operatorname{Cov}(F_a^2, G_a^2) &= 2 - 24a^2 + 56a^4. \end{aligned} \tag{23}$$

Subtracting $(1 + 2a^2)^2$ proves the second identity. The roots of the latter polynomial in a^2 are $(3 - \sqrt{2})/14$ and $(3 + \sqrt{2})/14$. Thus the square covariance is negative throughout the open interval between those values of a^2, despite both targets being centered. These parameter formulas are direct polynomial deductions from the displayed family; no additional general formal theorem is being claimed.

The mapped witness is $a = 1/2$, for which

$$F_{1/2} G_{1/2} = \tfrac{1}{2} - \tfrac{1}{4} H_4(Z), \qquad \|F_{1/2}\|_2^2 = \|G_{1/2}\|_2^2 = \tfrac{3}{2},$$

and hence

$$\mathbb{E}(F_{1/2}^2 G_{1/2}^2) = \tfrac{1}{4} + \tfrac{1}{16}\, 24 = \tfrac{7}{4} < \tfrac{9}{4}, \qquad \operatorname{Cov}(F_{1/2}^2, G_{1/2}^2) = -\tfrac{1}{2}.$$

Thus even the weaker inequality $\mathbb{E}(F^2 G^2) \geq \mathbb{E}F^2 \mathbb{E}G^2$ fails on centered mixed-chaos polynomials. This is neither a signed-measure effect nor failure of moment integrability. The measure is the same positive Gaussian product probability as in Theorem 1.1. At $a \neq 0$ both the first- and second-chaos projections of each target are nonzero; neither target belongs to a single homogeneous chaos. Consequently the counterexample does not contradict the homogeneous-chaos square inequalities of Malicet–Nourdin–Peccati–Poly [3], nor an inequality whose targets are jointly Gaussian linear variables. The mixed pair is a nonlinear polynomial image of a Gaussian variable, which is a different hypothesis.

13 Two skew factors and an actual determinant law

13.1 Symmetrize the combined coefficient, not its factors

For a finite real square matrix A, write $S_A = (A + A^T)/2$, $K_A = (A - A^T)/2$. For matrices A, B on possibly different finite index sets, the tensor coefficient acts on pairs of indices by $(A \otimes B)_{(i,a),(j,b)} = A_{ij}B_{ab}$. Its full symmetrization is

$$S_{A\otimes B} = S_A \otimes S_B + K_A \otimes K_B. \tag{24}$$

Indeed $(A \otimes B)^T = A^T \otimes B^T$. Substitute $A = S_A + K_A$, $A^T = S_A - K_A$ and the corresponding formulas for B: the two mixed terms cancel while the two displayed terms remain. This is an entrywise identity, with no positivity or commutation hypothesis. A symmetric factor tensored with a skew factor is fully skew and has zero diagonal quadratic contraction, but two skew factors give a fully symmetric tensor.

Take

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad T = J \otimes J.$$

Then $S_J = 0$, whereas $S_T = T \neq 0$: its quadratic contraction on four coordinates is

$$Q_T = 2(G_{00}G_{11} - G_{01}G_{10}). \tag{25}$$

Here the four G_{ia} are independent standard Gaussians and $\operatorname{tr} T = 0$, so this is exactly the centered observable (4). Also $T^T = T$, $T^2 = I_4$ and $\|T\|_{\mathrm{HS}}^2 = 4$, by $J^T = -J$ and $J^2 = -I_2$. Equations (5) and (6) give

$$\mathbb{E}Q_T^2 = 8, \qquad \mathbb{E}Q_T^4 = 12 \cdot 4^2 + 48 \cdot 4 = 384, \qquad \frac{\mathbb{E}Q_T^4}{(\mathbb{E}Q_T^2)^2} = 6.$$

Thus componentwise symmetrization would erase a genuinely nonzero second-chaos target. The single-factor skew field $V_J = JG$ has $\delta V_J = Q_J = 0$, while the combined field $V_T = TG$ satisfies

$$\delta V_T = Q_T, \qquad DQ_T = 2V_T, \qquad \|V_T\|_{L^2(H)}^2 = 4.$$

These are the original polynomial derivative/divergence identities of Section 3, now applied to the combined coefficient. This does not say that the ordinary product of two zero random variables is nonzero. Forming a tensor of coefficient matrices and then one quadratic observable is a different operation from multiplying the two individual quadratic observables.

13.2 The actual four-coordinate law, including arbitrary embeddings

For completeness the construction works with any four orthonormal vectors $h_{ia} \in \ell^2$ in the original countable Gaussian source. Define $W(h)$ by the L^2 limit of $\sum_{j<n} h_j G_j$. The four $W(h_{ia})$ have the independent standard Gaussian law. Here is a direct verification that retains the actual embedding. For any finite real coefficients t_{ia}, their linear combination is $W(\sum t_{ia}h_{ia})$. Its finite coordinate sums have characteristic function tending to $\exp(-\|\sum t_{ia}h_{ia}\|^2/2)$. The limit is valid since L^2 convergence implies L^1 convergence and $|e^{ix} - e^{iy}| \leq |x - y|$. Orthonormality makes the squared norm $\sum t_{ia}^2$. The joint characteristic function therefore factors into four standard Gaussian characteristic functions; uniqueness identifies the full product law. In particular the two rows are independent Gaussian vectors, not merely uncorrelated polynomial targets. Substitution in (25) transports the same law to the original ambient quadratic observable for every such frame.

Write $U = (G_{00}, G_{01})$, $V = (G_{10}, G_{11})$. Conditional on the first row $U = u$, (25) is the linear Gaussian combination $-2u_2V_1 + 2u_1V_2$. Hence, for real t,

$$\mathbb{E}(e^{itQ_T} \mid U = u) = e^{-2t^2(u_1^2+u_2^2)}. \tag{26}$$

The formula also holds at $u = 0$, where the conditional variable is zero. Every phase is bounded by one, so Fubini applies without a moment or uniform-integrability assumption. For a standard Gaussian Z and $c \geq 0$, combining its density with e^{-cZ^2} and scaling the Gaussian integral gives

$$\mathbb{E}e^{-cZ^2} = (1 + 2c)^{-1/2}.$$

Indeed the integral of $e^{-(c+1/2)x^2}$ divided by that of $e^{-x^2/2}$ has exactly this value by the substitution $y = \sqrt{1 + 2c}\, x$. Integrate (26) over the two independent first-row coordinates to obtain

$$\mathbb{E}e^{itQ_T} = \frac{1}{1 + 4t^2}. \tag{27}$$

This identifies the whole law, beyond the fourth-moment diagnostic.

The nonnegative density $p(x) = e^{-|x|/2}/4$ integrates to one. Its characteristic function is, by absolutely convergent half-line integration,

$$\int_{\mathbb{R}} e^{itx} p(x)\, dx = \frac{1}{4}\left(\frac{1}{1/2 - it} + \frac{1}{1/2 + it}\right) = \frac{1}{1 + 4t^2}.$$

For example the first half-line formula follows by integrating $e^{-(1/2-it)x}$ from zero to a finite endpoint and then letting that endpoint tend to infinity; the boundary exponential tends to zero in modulus. The negative half follows by $x \mapsto -x$. Characteristic-function uniqueness and (27) prove that Q_T has the actual Laplace density p. In particular its law has no atoms, is independent of the chosen orthonormal source frame, and the unscaled determinant has density $e^{-|x|}/2$. No higher-dimensional determinant law is asserted.

This distribution is a classical antecedent, not a new consequence claimed as a probability discovery. Nyquist–Rice–Riordan [4], page 98, explicitly display the unscaled two-dimensional Gaussian determinant density $e^{-|x|}/2$ and already call that special case well known. The formalization contribution is the exact combined-index symmetry, original derivative/divergence normalization, full law transport from arbitrary orthonormal source coordinates and direct density identification. No geometric or stochastic-process application follows solely from this tensor calibration.

14 Scope and formal correspondence

All sources are frozen at the verified commit

`bab688bb6140cd6316c80680095a914ea0d346db`.

The theorem map records exact declarations, full headers, source hashes and baseline links. The original full-build and complete transitive-axiom receipt and its referenced evidence hashes are checked separately. No research source was edited and no competing full research build was run.

The principal sources concern original linear-field divergence completion and recovery onto the second chaos, finite Gaussian matrix moments, strong box approximation, actual completed products, degree support, exact cumulants, strict contractions and mixed square covariance, together with the explicit centered mixed-chaos negative square-covariance witness, full combined tensor symmetry and transported determinant density. The general parameter interval for that witness is a

printed polynomial deduction. The symmetric square-summable matrix model above is a coordinate reconstruction of their original source-field unitary. The direct L^4 Cauchy proof reorganizes the finite sharp bound; the formal square construction instead first identifies an actual L^2 square through bounded approximation. Both retain the same actual target, original measure and operator normalization.

The paper's general completed statements use the projection conditions in (3). The matrix-annihilation independence statement is presented for finite symmetric matrices. No general infinite-dimensional independence classification, Gaussian approximation rate or nonlinear process identification is silently added. A Wick covariance power alone does not identify a fractional process, generator or stochastic calculus.

Lancaster's primary indexed Theorem 1 gives the classical trace-cumulant antecedent; its full theorem comparison remains incomplete. Ogawa's primary introductory centered theorem and finite orthogonal-range discussion supply the independence context. The formalization value here is the complete extension to original Gaussian L^2 targets and products with sharp constants, together with explicit nonattainment and retained contraction terms. None of these classical mechanisms is represented as a priority claim. Malicet–Nourdin–Peccati–Poly, Theorem 1.1 and its introductory two-variable discussion, provide a direct primary antecedent for homogeneous-chaos square-product nonnegativity, in much greater generality than needed for the second-chaos inequality here. Our exact contraction formula and explicit mixed-chaos boundary are a formalization exposition, not a new general product inequality.

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